

End-to-End System Architecture & Product Roadmap

Product	AppointUnified — Unified Smart Appointment Engine
Sectors	Healthcare · Government · Service Industry
Stack	Spring Boot · Next.js 14 · Neon PostgreSQL · Upstash Redis
Versions	V1 → V9 MVP to Enterprise
Date	April 03, 2026

■ Healthcare

■ Government

■ Services

This document covers the complete system design, version roadmap (V1–V9), all pages and UI elements per module, database schema, API contracts, tech stack decisions, free-tier tool usage, and CI/CD pipeline for AppointUnified — a production-grade multi-sector appointment and queue management platform.

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What Is AppointUnified?

AppointUnified is a Unified Smart Appointment & Queue Engine that serves three critical sectors — Healthcare, Government, and Service Industry — from a single platform. Unlike generic scheduling tools that are merely static calendars, AppointUnified is an adaptive scheduling system that survives real-world chaos: delays, no-shows, emergencies, and multi-party coordination.

Core Sector Problems

Healthcare	Government	Services
• Patients book 10 AM, seen at 12:30 PM • No-shows waste critical slots • Emergency patients break queue • No integration with labs/pharmacy • No priority for critical cases	• Queue lines despite online booking • Bot-blocked slots via agents • Corruption and manual overrides • Confusing government portals • Zero transparency on delays	• Providers show up late • Time estimates are unrealistic • Double bookings are common • Travel time completely ignored • Last-minute cancellations hit revenue

The Core Differentiator

Current Systems	AppointUnified
Static calendar — slot selected, done	Adaptive scheduling engine — survives chaos
No delay handling	Real-time dynamic rescheduling
Single sector only	Multi-sector from one platform
No trust/verification	4-layer trust system with badges
No queue intelligence	Live token queue with ETA

2. System Architecture (High Level)

Architecture Layers

Layer	Technology	Hosting
Frontend	Next.js 14 (SSR + App Router)	Vercel (free)
API Gateway	Next.js Server Actions + API Routes	Built-in
Backend	Spring Boot Microservices (Java 21)	Railway (free)
Cache/RT	Upstash Redis (WebSocket + Queue)	Upstash free
Database	Neon PostgreSQL (serverless)	Neon free
Storage	Cloudinary (images/docs)	Cloudinary free
Auth	Firebase Auth + Spring JWT	Firebase free
Notifications	SendGrid + FCM + WhatsApp API	Free tiers

Request Flow

1. User browser → Next.js SSR page (Vercel CDN)
2. Client action → Next.js Server Action (no CORS exposure)
3. Server Action → Spring Boot REST API (internal call, JWT attached)
4. Spring Boot → Neon PostgreSQL (query / write)
5. Spring Boot → Upstash Redis (queue update / cache)
6. Spring Boot → Notification Service (SendGrid / FCM / WhatsApp)
7. WebSocket channel → Upstash Redis pub/sub → Next.js client (live queue)
8. Cloudinary direct upload (presigned URL, never hits Spring Boot)

Spring Boot Microservices

Service	Responsibility
Auth Service	JWT / OAuth2 login, role management, refresh tokens
Booking Service	Slot creation, conflict detection, cancel/reschedule
Queue Engine	Real-time token generation, delay propagation, ETA calc
Notification Service	Email/SMS/push scheduler, WhatsApp Cloud API
Verification Service	KYC document intake, admin workflow, badge emission
Workflow Engine	Multi-step appointment chains (Doctor → Lab → Pharmacy)
Analytics Service	Aggregated metrics, no-show scoring, peak detection
Geo Service	OpenRouteService integration, travel time, buffer zones

3. Module Structure & Roles

Public User Module	Any citizen. Browse verified professionals across all 3 sectors, book appointments, track queue position, rate providers.
Healthcare Professional	Verified doctors, specialists, diagnostics labs. Manage schedules, accept/decline, view patient queue, complete appointments.
Government Professional	Verified government officers. Manage office slots, document validation queues, publish live status.
Service Professional	Verified technicians, salon staff, repair experts. Manage geo-schedule, travel buffer, service combos.
Healthcare Admin	Approve/reject doctor credentials, NMC verification, manage healthcare sector services and policies.
Government Admin	Verify officer employee IDs, manage public office slots, anti-bot monitoring, live status updates.
Services Admin	Verify service provider credentials, manage service catalogue, pricing rules, geo-coverage zones.
Super Admin	Full system audit logs, fraud detection, override any decision, platform analytics, user suspension.

Landing Page

- Hero section: 'Book any appointment, instantly' + CTA buttons (Book Now / Join as Professional)
- Sector selector cards: Healthcare / Government / Services with icons and stats
- Live stats strip: Total verified providers · Bookings today · Cities covered
- How It Works: 3-step animated explainer (Search → Book → Track)
- Featured Providers carousel with verified badges and ratings
- Testimonials section with sector-tagged reviews
- App download banner (PWA-ready)
- Footer: About, Terms, Privacy, Contact, Social links

Registration / Onboarding

- Role selector: 'I want to book' / 'I am a professional'
- Phone number entry with country code picker
- OTP verification input (6-digit, resend timer)
- Name, email, profile photo (optional at signup)
- Sector preference selection (which sectors you'll use)
- Terms & Privacy consent checkbox with link
- Skip option → guest booking (limited features)

Login Page

- Phone + OTP or Email + Password dual option
- Google / Firebase social login button
- Forgot password / resend OTP flow
- Remember device toggle
- 'Continue as guest' for quick browse

Sector Home (Healthcare / Government / Services)

- Search bar: search by professional name, specialty, or location
- Filter panel: Rating, Distance, Availability Today, Verified Only, Price Range
- Professional cards grid: Photo, Name, Verified badge, Rating (stars), Next available slot
- Sort options: Relevance / Distance / Rating / Price
- Map view toggle (Mapbox embed showing provider pins)
- Sector-specific category tabs: e.g., General / Specialist / Diagnostics for Healthcare
- Promotions strip: First-time booking offers

Professional Profile Page

- Cover + Avatar with verified badge and sector tag
- Name, Designation, License/Reg number (public)
- Star rating with review count, completion rate badge
- About section + qualifications + experience timeline
- Services offered table: Name / Duration / Price / Book button
- Availability calendar (read-only, shows open/booked slots)
- Reviews section: sorted by date, with like/flag buttons
- Report provider button (fraud protection)

- Share profile / Add to favorites

Booking Flow (Multi-step wizard)

- Step 1 — Select Service: service cards with duration + price
- Step 2 — Select Date: calendar with available dates highlighted
- Step 3 — Select Time Slot: time grid with open/booked visual
- Step 4 — Enter Details: name, phone, reason/notes (optional)
- Step 5 — Review & Confirm: summary card with total cost
- Step 6 — Payment (optional): Razorpay/Stripe integration
- Booking success screen: Appointment ID, QR code, add to calendar button
- WhatsApp / Email confirmation send option

My Bookings Dashboard

- Upcoming appointments tab: provider name, date/time, service, status pill
- Past appointments tab: completed / cancelled with timestamps
- Queue live tracker: 'You are #3 in queue · Est. 18 mins' with progress bar
- Cancel / Reschedule buttons per appointment
- Add to Google Calendar button
- Rate & Review prompt after completed appointment
- Waitlist entries: 'Joined waitlist for Dr. Sharma — Mon 3 PM'

Profile & Settings

- Avatar upload (Cloudinary)
- Personal details edit: name, email, phone, address
- Notification preferences: Email / SMS / WhatsApp / Push toggles
- Linked accounts: Google connected
- Booking history export (CSV)
- Account deactivation with confirmation
- Language & Region settings

Notification Center

- Bell icon in nav with unread count badge
- Full list: appointment reminders, queue updates, verification status, system alerts
- Mark all as read / delete actions
- Filter by type: Appointment / Queue / System

Professional Registration

- Role selection: Doctor / Government Officer / Service Technician
- Personal info: name, phone OTP, email
- Sector-specific credential fields:
 - Healthcare: NMC Registration number, Degree (MBBS/MD), Specialty, Hospital affiliation
 - Government: Employee ID, Department, Office address, official email (.gov.in)
 - Services: Trade/skill, years of experience, Aadhaar, service area
- Document upload (Cloudinary): certificate, ID proof, license scan
- Service area selection (map picker for service professionals)
- Terms + legal disclaimer: 'False credentials = account termination + legal action'
- Submission confirmation: 'Under review — typically 24-48 hours'

Verification Status Page

- Timeline view: Submitted → Under Review → Approved / Rejected
- Document checklist with status per item
- Admin notes / rejection reason (if rejected)
- Re-upload option for rejected documents
- Verified badge preview once approved

Professional Dashboard (Post-verification)

- Today's schedule: appointment cards in chronological order
- Queue status control: Start Queue / Pause Queue / Mark Delay (+N mins)
- Quick stats: Today's bookings / Completed / No-shows / Revenue
- Upcoming appointments calendar widget
- Pending reviews notification
- Profile completeness meter with suggestions

Schedule Management

- Weekly availability grid: toggle slots on/off per day
- Break/leave blocking: mark dates as unavailable
- Service duration settings per service type
- Buffer time configuration between appointments
- Override: add ad-hoc available slot outside normal hours
- Recurring pattern: set weekly schedule once

Appointment Management

- All appointments list with filters: Today / Week / Status
- Per-appointment actions: Confirm / Reschedule / Cancel / Mark Complete / Mark No-show
- Patient/client brief: name, reason for visit, history (Healthcare)
- Document checklist for government appointments
- Add notes to appointment (internal, not visible to user)
- Multi-step workflow progress (Healthcare: Doctor → Lab → Pharmacy)

Queue Control Panel (Live)

- Live token board: current serving number, waiting count
- Delay trigger: '+10 mins / +20 mins / Custom' — propagates to all waiting users
- Next patient call button (auto-notifies user)
- Emergency slot insertion (Healthcare priority override)
- Pause queue with reason (lunch / technical break)
- End-of-day summary auto-generated

Earnings & Analytics (V5+)

- Revenue chart: daily / weekly / monthly
- Service breakdown: which services earn most
- No-show rate trend
- Peak hours heatmap
- Payout history (if payment enabled)
- Ratings trend over time

Sector Admin Pages (Healthcare / Government / Services)

Admin Dashboard

- Pending verifications count with urgency flags
- Recent approvals / rejections activity feed
- Active professionals in sector
- Today's total bookings across sector
- Fraud alerts panel (duplicate docs, suspicious activity)
- Quick actions: Verify queue / Export report / Send announcement

Verification Queue

- List of pending professional applications with submission date
- Filter: Pending / Approved / Rejected / Flagged
- Per-application: view all uploaded documents, credential details
- Verification actions: Approve / Reject with reason / Flag for review
- Document hash check status (duplicate detection)
- Bulk actions: approve multiple low-risk submissions
- Audit trail: who approved/rejected, when, and why

Professional Management

- All professionals in sector with status pills
- Search/filter by name, status, registration number
- Edit professional profile (admin override)
- Suspend / Unsuspend with reason logging
- Re-verification trigger (periodic renewal enforcement)
- Complaint count per professional

Service Catalogue Management

- Add / edit / delete services in sector
- Set pricing ranges and duration defaults
- Assign services to professional categories
- Toggle service visibility (enable/disable)

Reports & Analytics

- Verification turnaround time report
- No-show rates by professional
- Complaint analysis: type and frequency
- Sector growth: new professionals per week
- Export all as CSV / PDF

Super Admin Pages

Super Admin Dashboard

- Platform-wide KPIs: total users, bookings, professionals, revenue
- Cross-sector health: which sector has most activity / issues
- Active fraud alerts from all sectors
- System health: API latency, DB connections, Redis queue depth

- Recent admin actions feed (all sector admins)

Audit Logs

- All system events: timestamp, actor, action, target, IP
- Filter by actor role, action type, date range
- Export audit log as CSV with tamper-evident hash
- Alert on unusual patterns (mass rejections, off-hours logins)

Fraud Detection Console

- Duplicate credential alerts: same doc hash used by multiple accounts
- Bot detection: accounts with bulk bookings from same IP
- Suspended account management
- Manual investigation workflow: flag → investigate → action
- IP block management

Platform Configuration

- Feature flags per sector (enable/disable features per version rollout)
- Notification template management
- Free tier limit monitoring: Neon connections, Upstash requests, SendGrid quota
- System announcements broadcast
- Maintenance mode toggle

Super Admin User Management

- All users across all roles and sectors
- Impersonate user (read-only, audit-logged)
- Force password reset / OTP re-verification
- Merge duplicate accounts
- Account deletion with data retention policy

7. Version Roadmap V1 → V9

V1 — Core Booking Engine

Features	New APIs	DB Tables	New Pages
• User signup/login (JWT)	• POST /auth/signup	• users	• Landing page
• Book time slot	• POST /auth/login	• professionals	• Register/Login
• Provider CRUD	• GET /slots	• appointments	• Sector home
• Basic booking page	• POST /appointments	• services	• Provider profile
• Confirmation email	• GET /providers		• Booking wizard
			• My Bookings

V2 — Multi-Sector + Trust System

Features	New APIs	DB Tables	New Pages
• Sector abstraction layer	• POST /verification/submit	• verifications	• Verification status page
• Document upload (Cloudinary)	• PATCH /verification/approve	• sectors	• Admin verification queue
• Admin verification workflow	• GET /professionals/verified	• documents	• Professional registration (enhanced)
• Verified badge on profile	• POST /auth/otp	• admin_actions	• Admin dashboard
• OTP verification			

V3 — Real-Time Queue System

Features	New APIs	DB Tables	New Pages
• Live token generation	• WS /queue/{apptId}	• queue_tokens	• Live queue tracker (user)
• WebSocket queue updates	• POST /queue/delay	• queue_events	• Queue control panel (professional)
• Queue position display	• GET /queue/status	• delay_log	• Token display page
• Upstash Redis pub/sub	• POST /queue/next		
• Delay notifications			

V4 — Dynamic Time Adjustment

Features	New APIs	DB Tables	New Pages
• Delay engine (propagates to all upcoming)	• POST /delay/trigger	• delay_events	• Delay notification toast (user)
• Auto-reschedule cascade	• GET /appointments/eta	• reschedule_log	• Delay trigger UI (professional)
• ETA recalculation	• POST /emergency/insert		• Emergency queue insertion
• Push notification on delay	• PUT /appointments/reschedule-cascade		
• Emergency slot insertion			

V5 — No-Show Prediction

Features	New APIs	DB Tables	New Pages
- Behavior scoring engine - Cancellation history tracking - Risk flag on high-risk bookings - Overbooking strategy - Waitlist auto-notification	- GET /users/{id}/risk-score - POST /waitlist/join - GET /waitlist/{slotId} - POST /waitlist/notify	- behavior_scores - cancellation_history - waitlist	- Risk indicator (admin view) - Waitlist join button - Overbooking config (professional)

V6 — Multi-Step Workflow Engine

Features	New APIs	DB Tables	New Pages
- Workflow definition (JSON steps) - Auto-next appointment trigger - Cross-provider chain (Doctor→Lab) - Step completion tracking - Linked appointment view	- POST /workflows - GET /workflows/{id}/steps - POST /workflows/{id}/complete-step - GET /appointments/{id}/workflow	- workflows - workflow_steps - workflow_instances	- Workflow progress tracker (user) - Workflow builder (admin) - Multi-step appointment view

V7 — Geo-Aware Scheduling

Features	New APIs	DB Tables	New Pages
- OpenRouteService travel time API - Buffer time between appointments - Provider route optimization - Live technician tracking - Geo-zone coverage management	- GET /geo/travel-time - GET /providers/nearby - GET /tracking/{providerId} - POST /geo/optimize-route	- geo_zones - provider_locations - route_cache	- Map view (user) - Provider location tracker - Service area editor (professional)

V8 — Smart Priority Engine

Features	New APIs	DB Tables	New Pages
- Priority tiers: Emergency > Premium > Normal - Queue jumping logic - Fairness balancing algorithm - Priority booking UI - SLA enforcement	- POST /appointments/priority - GET /queue/priority-check - POST /priority/override	- priority_rules - sla_configs	- Priority badge on booking - Emergency booking flow - Priority queue view (professional)

V9 — Analytics Intelligence

Features	New APIs	DB Tables	New Pages
• Admin analytics dashboard	• GET /analytics/summary	• analytics_snapshots	• Analytics dashboard (admin)
• No-show prediction charts	• GET /analytics/no-show-trends	• report_exports	• Super admin KPI board
• Peak hour heatmap	• GET /analytics/peak-hours		• Professional earnings page
• Revenue analytics	• GET /analytics/revenue		• Export center
• Slot optimization suggestions			
• Export reports (PDF/CSV)			

8. Database Schema (Neon PostgreSQL)

Table: users

Column	Definition
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
email	VARCHAR(255) UNIQUE
phone	VARCHAR(20) UNIQUE NOT NULL
password_hash	TEXT
role	ENUM('PUBLIC','PROFESSIONAL','ADMIN','SUPER_ADMIN')
sector	ENUM('HEALTHCARE','GOVERNMENT','SERVICES') NULLABLE
is_verified	BOOLEAN DEFAULT FALSE
risk_score	DECIMAL(3,2) DEFAULT 0.0
created_at	TIMESTAMPTZ DEFAULT NOW()
updated_at	TIMESTAMPTZ

Table: professionals

Column	Definition
id	UUID PRIMARY KEY
user_id	UUID REFERENCES users(id)
display_name	VARCHAR(255)
sector	ENUM('HEALTHCARE','GOVERNMENT','SERVICES')
specialty	VARCHAR(255)
license_number	VARCHAR(100) UNIQUE
license_hash	TEXT (tamper detection)
verification_status	ENUM('PENDING','APPROVED','REJECTED','SUSPENDED')
rating_avg	DECIMAL(3,2)
total_reviews	INTEGER DEFAULT 0
avatar_url	TEXT (Cloudinary URL)
bio	TEXT
service_area_radius_km	DECIMAL(6,2) NULLABLE
created_at	TIMESTAMPTZ DEFAULT NOW()

Table: services

Column	Definition
id	UUID PRIMARY KEY
sector	ENUM
name	VARCHAR(255)
description	TEXT
duration_minutes	INTEGER

price	DECIMAL(10,2)
is_active	BOOLEAN DEFAULT TRUE
created_by_admin	UUID REFERENCES users(id)

Table: appointments

Column	Definition
id	UUID PRIMARY KEY
client_id	UUID REFERENCES users(id)
professional_id	UUID REFERENCES professionals(id)
service_id	UUID REFERENCES services(id)
start_time	TIMESTAMPTZ NOT NULL
end_time	TIMESTAMPTZ NOT NULL
status	ENUM('SCHEDULED','IN_QUEUE','IN_PROGRESS','COMPLETED','CANCELLED','NO_SHOW')
priority	ENUM('NORMAL','PREMIUM','EMERGENCY') DEFAULT 'NORMAL'
notes	TEXT
is_virtual	BOOLEAN DEFAULT FALSE
meet_link	TEXT NULLABLE
workflow_instance_id	UUID NULLABLE

Table: queue_tokens

Column	Definition
id	UUID PRIMARY KEY
appointment_id	UUID REFERENCES appointments(id)
token_number	INTEGER
position	INTEGER
estimated_wait_mins	INTEGER
called_at	TIMESTAMPTZ NULLABLE
created_at	TIMESTAMPTZ DEFAULT NOW()

Table: availability

Column	Definition
id	UUID PRIMARY KEY
professional_id	UUID REFERENCES professionals(id)
weekday	SMALLINT (0=Mon, 6=Sun)
start_time	TIME
end_time	TIME
buffer_minutes	INTEGER DEFAULT 5
is_active	BOOLEAN DEFAULT TRUE

Table: verifications

Column	Definition
id	UUID PRIMARY KEY
professional_id	UUID REFERENCES professionals(id)
doc_type	ENUM('LICENSE','DEGREE','ID_PROOF','EMPLOYEE_ID','EXPERIENCE')
doc_url	TEXT (Cloudinary)
doc_hash	TEXT (SHA-256)
status	ENUM('PENDING','APPROVED','REJECTED')
reviewed_by	UUID REFERENCES users(id) NULLABLE
review_notes	TEXT
reviewed_at	TIMESTAMPTZ
submitted_at	TIMESTAMPTZ DEFAULT NOW()

Table: workflows

Column	Definition
id	UUID PRIMARY KEY
name	VARCHAR(255) e.g. 'Doctor → Lab → Pharmacy'
sector	ENUM
steps	JSONB [{step, service_id, order, auto_book}]
is_active	BOOLEAN DEFAULT TRUE

Table: reviews

Column	Definition
id	UUID PRIMARY KEY
appointment_id	UUID REFERENCES appointments(id)
reviewer_id	UUID REFERENCES users(id)
professional_id	UUID REFERENCES professionals(id)
rating	SMALLINT CHECK (rating BETWEEN 1 AND 5)
comment	TEXT
is_flagged	BOOLEAN DEFAULT FALSE
created_at	TIMESTAMPTZ DEFAULT NOW()

Table: audit_logs

Column	Definition
id	UUID PRIMARY KEY
actor_id	UUID REFERENCES users(id)
action	VARCHAR(100) e.g. 'APPROVE_PROFESSIONAL'
target_type	VARCHAR(50) e.g. 'professional'
target_id	UUID

metadata	JSONB
ip_address	INET
created_at	TIMESTAMPTZ DEFAULT NOW()

9. API Contract Reference

Auth APIs

Method	Endpoint	Description	Auth
POST	/api/auth/signup	Register user with phone OTP	PUBLIC
POST	/api/auth/login	Login, returns JWT access + refresh token	PUBLIC
POST	/api/auth/otp/send	Send OTP to phone	PUBLIC
POST	/api/auth/otp/verify	Verify OTP, returns temp token	PUBLIC
POST	/api/auth/refresh	Refresh JWT using refresh token	AUTH
DELETE	/api/auth/logout	Invalidate refresh token	AUTH

Professional APIs

Method	Endpoint	Description	Auth
POST	/api/professionals/register	Submit professional registration	AUTH
GET	/api/professionals	List verified professionals	PUBLIC
GET	/api/professionals/{id}	Get professional profile	PUBLIC
PATCH	/api/professionals/{id}	Update own profile	PROFESSIONAL
GET	/api/professionals/{id}/slots	Get available slots for a date	PUBLIC
POST	/api/professionals/{id}/schedule	Set weekly availability	PROFESSIONAL

Verification APIs

Method	Endpoint	Description	Auth
POST	/api/verification/submit	Upload documents for review	PROFESSIONAL
GET	/api/verification/status	Check own verification status	PROFESSIONAL
GET	/api/admin/verifications	List pending verifications	ADMIN
PATCH	/api/admin/verifications/{id}	Approve or reject with notes	ADMIN

Booking & Appointment APIs

Method	Endpoint	Description	Auth
POST	/api/appointments	Create appointment	AUTH
GET	/api/appointments/me	My upcoming appointments	AUTH
PUT	/api/appointments/{id}/cancel	Cancel appointment	AUTH
PUT	/api/appointments/{id}/reschedule	Reschedule appointment	AUTH
POST	/api/appointments/{id}/complete	Mark as completed	PROFESSIONAL
POST	/api/appointments/{id}/no-show	Mark as no-show	PROFESSIONAL

Queue APIs

Method	Endpoint	Description	Auth
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WS	/ws/queue/{apptId}	Subscribe to live queue updates	AUTH
POST	/api/queue/delay	Trigger delay for all upcoming (mins)	PROFESSIONAL
POST	/api/queue/next	Call next patient/client	PROFESSIONAL
GET	/api/queue/{profId}/status	Get live queue board	PUBLIC
POST	/api/queue/pause	Pause queue with reason	PROFESSIONAL

Analytics APIs

Method	Endpoint	Description	Auth
GET	/api/analytics/summary	Platform-wide KPIs	SUPER_ADMIN
GET	/api/analytics/no-show-trends	No-show rate per professional	ADMIN
GET	/api/analytics/peak-hours	Peak booking heatmap data	ADMIN
GET	/api/analytics/revenue	Revenue breakdown	ADMIN/PROFESSIONAL

10. Free Tier Tech Stack & Smart Usage

Tool	Purpose	Free Limit	Smart Usage Strategy
Vercel	Frontend hosting	100GB bandwidth, unlimited deploys	Use ISR for provider profile pages — cache for 60s, not static
Railway	Spring Boot hosting	500 hours/month free, 512MB RAM	Use sleep mode for dev env; keep prod always-on with \$5 plan if needed
Neon DB	PostgreSQL	0.5 GB storage, 1 branch, auto-scale to zero	Use connection pooling (PgBouncer built-in). Index queue_tokens and appointments heavily.
Upstash Redis	Cache + pub/sub	10,000 commands/day free	Only cache slot availability + queue position. TTL 30s. Batch pub/sub.
Firebase Auth	Authentication	10,000 users free	Use for phone OTP only. JWT handled by Spring Boot for API auth.
Firebase FCM	Push notifications	Unlimited push (free)	Use for appointment reminders. Batch sends to reduce API calls.
SendGrid	Email	100 emails/day free	Use for confirmations and reminders only. Queue with Spring Scheduler.
Cloudinary	Image/doc storage	25 credits/month free	Use for profile photos + document uploads. Enforce 2MB limit.
OpenRouteService	Maps/routing	2,000 requests/day free	Cache travel time results in Redis (TTL 1 hour). Only call on booking.
WhatsApp Cloud API	WhatsApp alerts	1,000 conversations/month free	Use only for critical alerts: queue updates, confirmation, emergency.
GitHub Actions	CI/CD	2,000 minutes/month free	Run tests on PR, deploy on merge to main only.
Resend	Developer email	3,000 emails/month free	Backup to SendGrid; use for transactional emails with better DX.

Free Tier Hard Limits — Plan Around These

- Neon DB auto-pauses after 5 minutes of inactivity. First query after pause has ~500ms cold start. Use a keep-alive ping every 4 minutes in production.
- Upstash 10K commands/day = ~7 per minute. Every queue subscription WebSocket message = 1 command. Batch updates instead of per-user push.
- Railway free tier has 512MB RAM. Spring Boot alone uses ~300MB. Profile memory at V3 before adding more services.
- SendGrid 100/day is gone fast. Queue emails in DB first, send via scheduled job. Never send on every API call.
- WhatsApp Cloud API free tier resets monthly. Track usage per user. Only send if user opted in.
- Firebase Auth 10K users is enough for MVP. After that, \$0.0055/user. Budget for this at V3.

11. CI/CD Pipeline

GitHub Actions Workflow Structure

Two workflow files live in `.github/workflows/`: one for CI (runs on every push/PR) and one for deployment (runs on merge to main).

`ci.yml`

```
# .github/workflows/ci.yml
name: CI
on: [push, pull_request]
jobs:
  nextjs-ci:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
    with: { node-version: '20' }
      - run: npm ci
      - run: npm run lint
      - run: npm run type-check
      - run: npm run test

  springboot-ci:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-java@v4
    with: { java-version: '21', distribution: 'temurin' }
      - name: Cache Maven
    uses: actions/cache@v3
    with:
      path: ~/.m2
    key: ${{ runner.os }}-maven-${{ hashFiles('**/pom.xml') }}
      - run: mvn -B clean test
      - run: mvn -B verify -DskipTests
```

`deploy.yml`

```
# .github/workflows/deploy.yml
name: Deploy
on:
  push:
    branches: [main]
jobs:
  deploy-frontend:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
    # Vercel auto-deploys via GitHub integration – no extra steps needed

  deploy-backend:
    runs-on: ubuntu-latest
    needs: [springboot-ci]
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-java@v4
    with: { java-version: '21', distribution: 'temurin' }
      - run: mvn spring-boot:build-image -DskipTests
      - name: Push to Docker Hub
    run: |
      echo ${ secrets.DOCKER_PASSWORD } | docker login -u ${ secrets.DOCKER_USERNAME } --password-stdin
      docker push yourusername/appointunified:latest
```

```
- name: Deploy to Railway
run: railway up --service backend
env:
RAILWAY_TOKEN: ${ secrets.RAILWAY_TOKEN }
```

Environment Variables (GitHub Secrets)

Variable	Where Used	Description
NEON_DATABASE_URL	Spring Boot	Full Neon connection string with SSL
JWT_SECRET	Spring Boot	64-char random string for JWT signing
UPSTASH_REDIS_URL	Spring Boot	Upstash REST URL for Redis
FIREBASE_CREDENTIALS	Spring Boot	Firebase Admin SDK JSON (base64 encoded)
SENDGRID_API_KEY	Spring Boot	SendGrid API key for email sending
CLOUDINARY_URL	Spring Boot + Next.js	Cloudinary upload URL with API secret
NEXT_PUBLIC_API_URL	Next.js	Spring Boot backend URL (Railway)
RAILWAY_TOKEN	GitHub Actions	Railway deployment token

13. Trust & Verification System

4-Layer Trust Architecture

Layer 1	Identity Verification (KYC)	• Phone OTP via Firebase Auth (mandatory) • Name + email verification • Government ID upload (Aadhaar/PAN for India) • Liveness check (V4+): selfie match
Layer 2	Credential Verification	• Healthcare: NMC Registration Number + Degree certificate • Government: Employee ID + official .gov.in email • Services: Aadhaar + experience proof + trade certification • Document SHA-256 hash stored to detect tampering
Layer 3	Admin Review (2-step)	• Automated check: duplicate hash detection, format validation • Manual admin review: 3 separate domain admins • Approval criteria: name match, valid registration, no duplicates • Rejection with reason + re-submission option
Layer 4	Continuous Trust Monitoring	• Post-booking ratings and reviews • Complaint system with escalation to Super Admin • No-show rate tracking per professional • Re-verification trigger every 12 months • Activity anomaly detection (sudden cancellation spikes)

14. Real-Time Queue Architecture

How the Queue Engine Works

Appointment Created

User books → appointment saved in Neon DB → queue_token row created with position N

WebSocket Subscribe

User's browser connects to WS endpoint /ws/queue/{apptId} → Spring Boot STOMP channel

Redis Pub/Sub

Queue state stored in Upstash Redis. Key: queue:{professionalId}. Value: JSON of all active tokens + positions

Professional Calls Next

POST /api/queue/next → Redis pops token #1 → broadcasts update to all subscribers via STOMP

Delay Trigger

Professional triggers +N mins delay → Spring Boot recalculates all ETAs → Redis updates → all subscribers receive new ETA

Emergency Insert

Emergency appointment created with priority=EMERGENCY → inserted at position 1 → queue reshuffled → all notified

Completion

Appointment marked COMPLETED → token removed from Redis → next token called → WhatsApp notification sent to next user

Queue Data in Redis

Key structure in Upstash Redis:

```
queue:{professionalId} → JSON {  
  "current_token": 5,  
  "delay_minutes": 20,  
  "tokens": [  
    {"id": "uuid", "appointment_id": "uuid", "position": 1, "eta": "14:35"},  
    {"id": "uuid", "appointment_id": "uuid", "position": 2, "eta": "15:05"},  
  ]  
}  
TTL: Set to end-of-day (auto-expires at midnight)
```

15. Execution Timeline

Phase 1: Foundation (Week 1–2)

Covers: V1 + V2

- Set up Neon DB, run Flyway migrations for all core tables
- Spring Boot: Auth service (JWT), Professional CRUD, Booking CRUD
- Next.js: Landing, Register/Login, Sector Home, Provider Profile, Booking Wizard
- Admin: Verification queue + approve/reject flow
- Deploy: Vercel (frontend) + Railway (backend) + GitHub Actions CI
- Goal: Working booking system with verification in production

Phase 2: Real-Time Core (Week 3–4)

Covers: V3 + V4

- Add Upstash Redis, Spring STOMP WebSockets
- Build queue token system end-to-end
- Live queue tracker page (user) + Queue control panel (professional)
- Delay engine + ETA recalculation + cascade reschedule
- WhatsApp Cloud API integration for queue updates
- Goal: Real differentiator — live queue like Swiggy tracking

Phase 3: Intelligence (Week 5–6)

Covers: V5 + V6

- Behavior tracking tables + risk scoring logic in Spring Boot
- Waitlist system + auto-notification on slot open
- Workflow engine: define multi-step flows in JSONB
- Auto-next appointment trigger on step completion (Healthcare)
- Goal: Smart system that learns and chains appointments

Phase 4: Geo + Priority (Week 7–8)

Covers: V7 + V8

- OpenRouteService integration for travel time
- Buffer time config per professional
- Mapbox embed on sector home (provider map view)
- Priority engine: EMERGENCY > PREMIUM > NORMAL queue logic
- SLA enforcement on premium bookings
- Goal: Production-grade for all 3 sectors end-to-end

Phase 5: Analytics + Polish (Week 9–10)

Covers: V9 + Hardening

- Analytics dashboard for admin (Recharts in Next.js)
- Super admin KPI board + fraud console
- Professional earnings page
- Performance: DB indexing audit, Redis batching, image optimization
- Security: Rate limiting, reCAPTCHA on booking, SQL injection audit
- Goal: Investor-ready / SaaS-ready product

AppointUnified — Built to survive real-world unpredictability. Not just a calendar. A dynamic scheduling system for Healthcare, Government, and Services. Stack: Spring Boot · Next.js 14 · Neon PostgreSQL · Upstash Redis · Vercel · Railway.